No of additional sheets 5

LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY

LOURDES HILLS, KUTTICHAL P. O., THIRUVANANTHAPURAMM - 695 574

(23)

ANSWER BOOK

Name Abin Thomas Class No. 3 Branch MCA

Subject Data Science and Machine Learning Semester : 3rd

Mark Tabulation

		Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	Q.14	Q.15
Mark Divisions		2	3	3	3	3	3		6							
Total																
Grand Total														23		

(1) Data Science is a collection of techniques used to extract meaningful and novel information in the form of pattern, or as information from a given set of dataset that all, further useful for other applications. Artificial intelligence, machine learning and data Science are all interrelated.

Data Science has many use in modern days. it can be used for medical purpose for some understanding understanding the Diseases by examining the Symptoms. it can also be used to find relation b/w two attributes

Features of Data science

2 i) Extracting meaningful patterns
ii) Building representative models. Examples

2) Machine learning is the process of introducing the to a machine using some set of training data and Machines understand things through experiences. So that the performance improves on similar experience in the future

Machine learning is divided into three related components

Examples diagram

1) Data storage

Here we identify data set used for learning and store it in computer

2) Abstraction

In this stage we assign meaning to stored data

eg: Acceleration due to

grape (Sweetness = 8 , Crunchiness = 5)

3) Generalization

we may find out lot of patter during abstra
stage. in this stage. limit these patters.
are relevent problem only

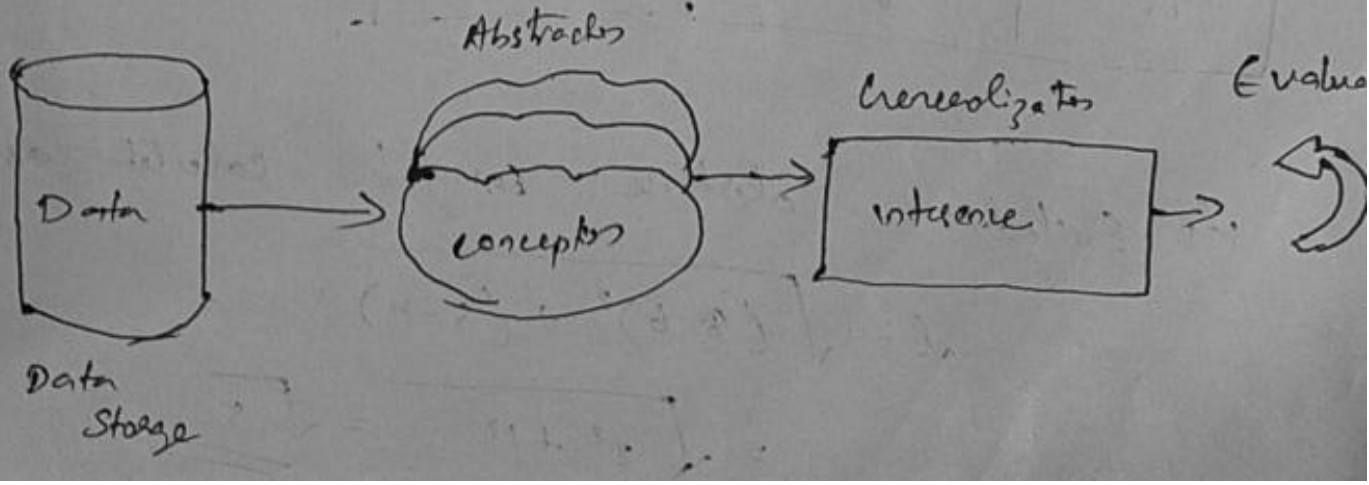
4) Evaluation

4) ~~Evaluation~~ Here we evaluate model using
data.

Types of Algorithm

1. Predictive model
2. Descriptive model
3. meta learning model

2 Diagram



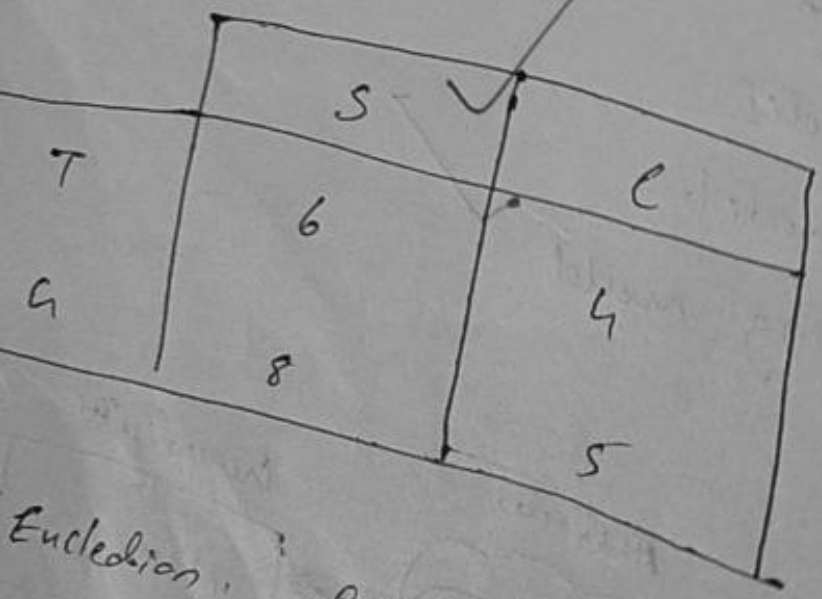
(Soccer = 8, Cricket = 5)

Euclidean distance is the distance between two points. It is to be measured by a ruler.

It is found using the formula

~~$d(P,Q) = \sqrt{(x_2 - x_1)^2}$~~

$$d(P,Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + \ldots + (z_2 - z_1)^2}$$



Euclidean distance can be calculated as

$$d = \sqrt{(8-6)^2 + (5-4)^2}$$

$$= \sqrt{}$$

(4) Bayes Theorm gives the conditional probability of an event A given another event B has occurred

its formula is

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B/A) \cdot P(A)}{P(B)}$$

$P(A/B)$:- Conditional probability of A given B

$P(B/A)$:- Conditional probabilty of B given A

$P(A)$:- probability of event A

$P(B)$:- probability of event B

Eg: ~~Rolling two dice~~

eg: Tossing a coin

Sample spaces = {HH, HT, TH, TT}

Head - H

Tails - T

$$P(A/B) = \frac{P(B/A)\ P(A)}{P(B)}$$

$$= \frac{1/2 \cdot 1/2}{1/2} = 1/2 = 0.5$$

(5) Classification with nearest neighbour Algorithm

The classification is done using Euclidian distance formulas.

K - no of neighbours taken

Euclidian distance b/w nearest neighbour is taken as the summarization is done with the corresponding data that has smallest distance from a the item we want

K is chosen with some criteria. usually K is chosen chosen as the square root of number of data items.

Eg: To classify tomato is a vegitable or a fruit.

with tomato has sweetness = 6 and

ingredients	Sweetness	Crunchiness	food type
grape	8	5	Fruit
orange	6	6	fruit
beans	3	2	vegetable

Euclidean distance

grape and tomato

$$= \sqrt{(8-6)^2 + (4-5)^2} = \sqrt{5}$$

grape and orange

$$= \sqrt{(6-6)^2 + (4-6)^2} = \sqrt{4} = 2$$

beans and orange

$$= \sqrt{(3-6)^2 + (4-2)^2} = \sqrt{13}$$

The shared attribute b/w tomato and orange so food type of tomato is fruit.

(6) Data Science problems are broadly classified into two types

Supervised learning model and unsupervised learning model

=) Supervised learning

- In Supervised learning prior knowledge about input and output data is given
- In this model training is given to the machine to get certain kind of output
- It is high accurate
- Knn, Naive Byes etc are supervised learning

input labeled data

x (features)	ylabel
x_{11} x_{12} .. x_{1n}	y_1
	⋮
x_{k1} x_{k2} .. x_{kn}	y_k

⇒ unsupervised learning

- no training data model is given
- it finds relations b/w two attributes in a data set
- less accurate
- complex model
- ~~no of clusters is predefined~~
- no o/p is produced

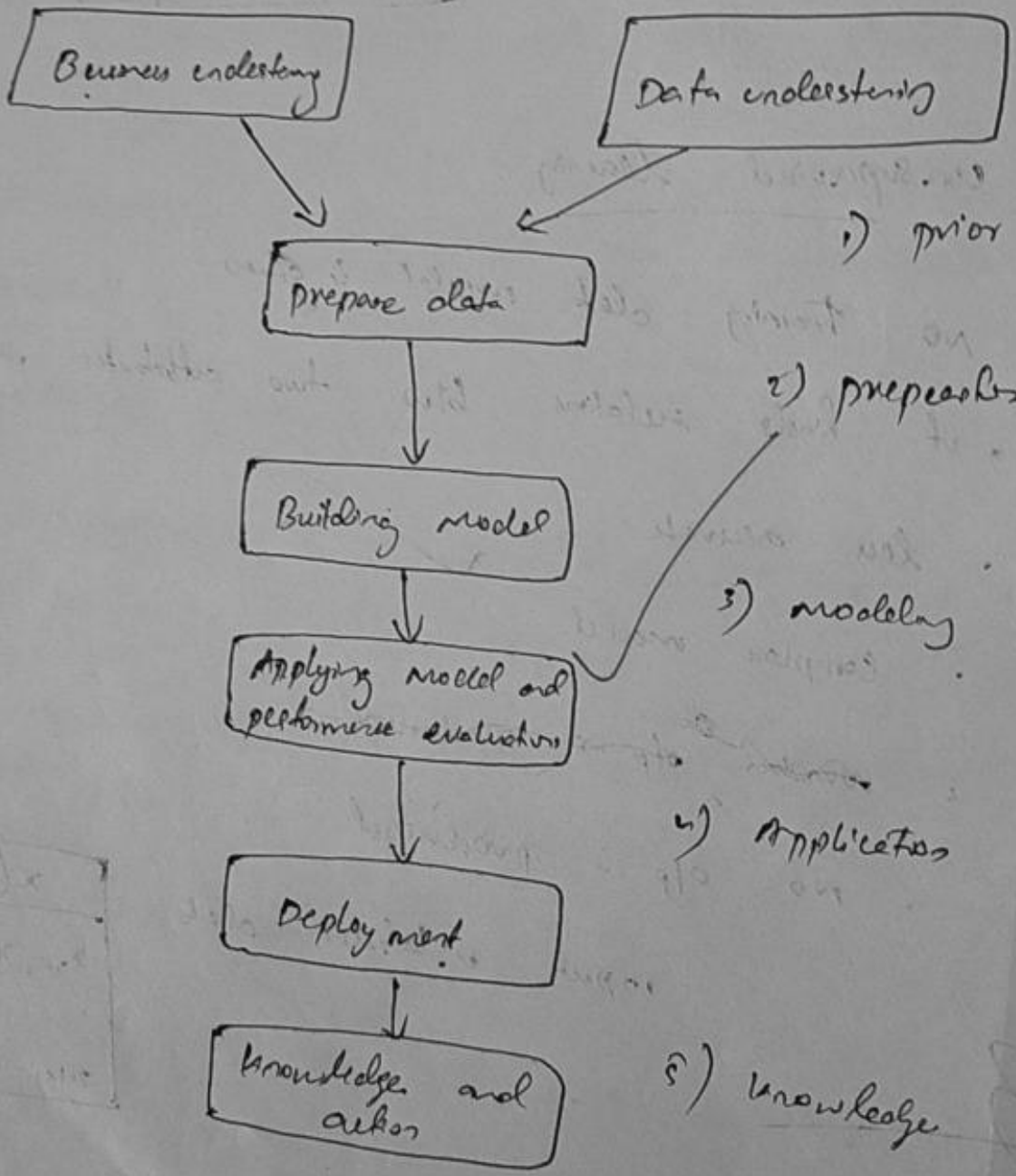
input unlabeled data

x (features)
x_{11}, x_{12}, ... x_{1n}
x_{k1} x_{k2} .. x_{kn}

3

(8)

process of preparing a dataset to perform a data science task



1) prior

2) preprocess

3) modeling

4) Application

5) knowledge

There are various process for preparing a dataset to perform a data science task.

i) prior knowledge

prior knowledge refers to information that is already known about the subject

objective: without a well defined statement we cannot solve the problem

Subject area: we should have knowledge about the subject that we all dealing with

Data set: A collection of data

Data point: A single instance in a dataset

Attribute: A single feature or property in a dataset

identifier: A special attribute that is used to locate a data

ii) Data preparation

Preparation a dataset to suit a data science task is the

most time consuming tasks, it involves.

Data exploration: Exploring the data to get the that we require

Data quality: Quality of the data is very important quality improves the correctness of data

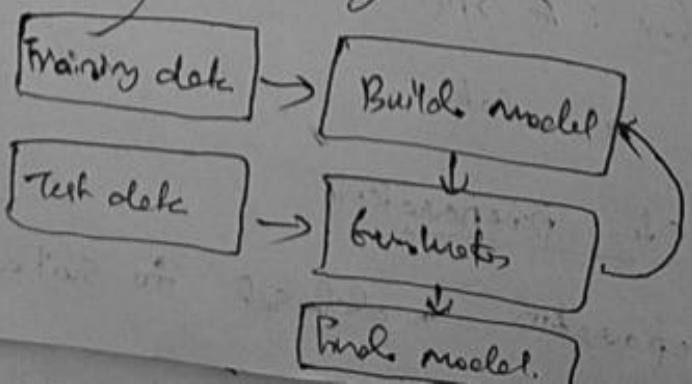
missing values: it is very troublesome to have missing values and it is to be substituted with some other suitable data

Data type and conversion: All the data should be of same type so the people and all are changes to given using specific parameters

other tasks are, ~~Transform~~ Transformation, outliers, feature selection are Data Sampling.

3) Modeling

A model is the Abstract representation of the data and the relationship is given data set



LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY

LOURDES HILLS, KUTTICHAL P. O., THIRUVANANTHAPURAMM - 695 574

ANSWER BOOK

Name..... Class No..... Branch.....

Subject..... Semester :.....

Mark Tabulation															
		Q. 1	Q. 2	Q. 3	Q. 4	Q. 5	Q. 6	Q. 7	Q. 8	Q. 9	Q. 10	Q. 11	Q. 12	Q. 13	Q. 14	Q. 15
Mark Divisions																
Total																
Grand Total																

Training and testing dataset. A data model used to create with known attributes is called training dataset while of the created model is tested using testing da

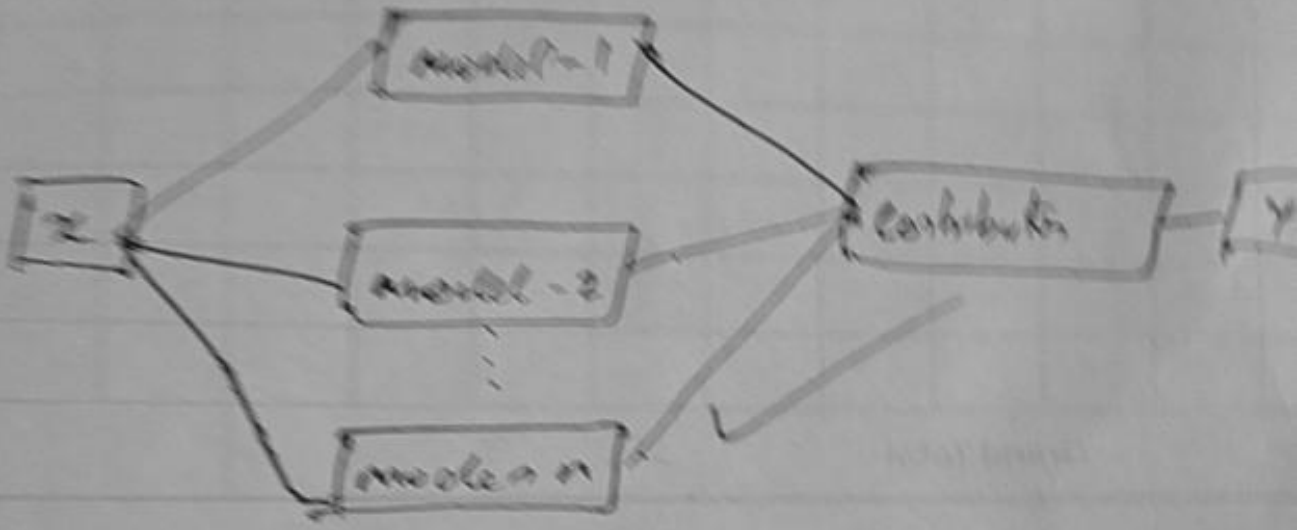
~~(a) Applications~~

Learning Algorithm :

It is by using algorithms that we train data, th
help to the practitioners to select the algorithm nec

Evaluation of model

The created model is evaluating test data ..



) Applicat

Deployment is the stage at which the
prediction ready or leave

Readiness

:

that determines

deployment

Response time : it is the time taken by the algorithm to give best output

model Refresh :

the model is not the most modern it is to be made aggressive update as technology increases it is also a need to update the model now and then

3) Knowledge

the data science provides a framework to extract non trivial information from data. To extract knowledge from those massive data assets advanced approaches needed to be employed like data science algorithms that science provides various options in terms of algorithms using these options to extract the right information from the data.